

MEng Thesis Proposal

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Abstract

As large language models (LLMs) become widely used, a key problem is how to align their outputs with human preferences. Current reward models used in reinforcement learning from human feedback (RLHF) are typically Bradley–Terry–style network, which often limits their robustness and ability to generalize to new prompts, users, and domains. I propose that this thesis will explore alternative reward-modeling approaches; some preliminary work done so far include combining embedding representations with matrix estimation and latent-factor techniques inspired by recommender systems. I test four methods—a Simple Matrix Estimation model, a Pairwise-to-Response model, a Score-Difference matrix factorization model, and a Difference-Vector Baseline model—which aim to impute user–response score matrices and leverage structure to generalize better across datasets. These methods are evaluated across several public human-feedback datasets, with a focus on pairwise accuracy, data efficiency, and out-of-distribution behavior. Preliminary results indicate mixed success, highlighting a further need to explore this space. This proposal also outlines future exploration, including embedding preprocessing and calibration, integration of chain-of-thought or process-based supervision, and modeling domain-specific user clusters. The overall goal of this research is to better understand how structured matrix methods can lead to better reward models for LLM alignment.

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1 Introduction

There is an increasing prevalence of large language models (LLMs) and other generative systems in today’s society, with these models being able to achieve impressive performance on a wide variety of tasks, ranging from dialogue to summarization to creative writing to code generation [7, 21]. However, with this rise, there remain many open problems on the best way to align these systems with human preferences. Raw pre-training or supervised fine-tuning often fails to align model outputs with these human values; what is syntactically valid or high-likelihood does not guarantee the output is helpful, safe, or aligned with user intent [2, 5, 11].

A growing and promising approach to address this challenge is to learn a reward model from human feedback, where the reward model essentially estimates how much a given (prompt, response) pair satisfies the human quality criteria. This reward model then can be used to tune the LLM (for instance through reinforcement learning) toward outputs that better match human preferences. This framework, often broadly referred to as *Reinforcement Learning from Human Feedback (RLHF)*, has become a standard tool for aligning LLMs with human intent [9, 23].

Traditionally, these reward models in RLHF rely on probabilistic pairwise comparison frameworks, especially variants of the **Bradley-Terry** (BT) model to infer latent “quality” scores for items from human preference data [6]. In these reward models, a Bradley Terry loss is used to update a pretrained LLM for supervised learning; during training, human preference data in the form of (prompt, chosen response, rejected response) and pairs are embedded and then used with a logistic loss to predict higher scores for responses humans prefer [26]. Despite this flexibility, embedding-based BT models still face limitations—the training data comes from a sparse subset of human comparisons, and the model does not explicitly encode latent structure across responses or users. This raises challenges in generalization, robustness to out-of-distribution prompts/responses, and personalization [3, 8, ?].

Thus, the goal of this thesis is to explore reward-modeling strategies that combine human feedback, embedding representations, and latent-factor or matrix-based methods. We focus on a few key questions as follows. First, we explore the preference data in RLHF, and determine if it can be modeled directly or induced into being able to be modeled as a low-rank or sparse + low-rank structure. Second, we explore whether our matrix imputation methods can outperform BT models in out-of-distribution generalization. Third, we explore

whether imposing this matrix structure reduces reward hacking during RL optimization. Overall, our goal is to explore the structure of the data and better capture relationships among prompts, responses, and users to create a human model that is able to be used in the downstream RLHF pipeline.

2 Related Work

2.1 Bradley-Terry Reward Modeling

Reward modeling is a central component of aligning large language models (LLMs) with human preferences under RLHF. A widely used approach is based on the probabilistic framework of the **Bradley–Terry (BT)** model [6], adapted for modern neural networks.

In practice, the reward model is implemented by taking a pretrained LLM and appending a scalar output head [26]. For any prompt–response pair (x, y) , the network outputs a real-valued score $r_\theta(x, y)$. Given human preference data as pairs (y_w, y_ℓ) , with one response being "better" than the other, the model is trained using a BT-style (logistic) loss so that $r_\theta(x, y_w) > r_\theta(x, y_\ell)$ is more likely. Concretely:

$$P(y_w \succ y_\ell \mid x) = \frac{\exp(r_\theta(x, y_w))}{\exp(r_\theta(x, y_w)) + \exp(r_\theta(x, y_\ell))}$$

and training minimizes the negative log-likelihood (cross-entropy) over the preference dataset. This embedding-based BT approach supports scoring any (x, y) , which we can then interpret as the "quality" of the responses. If the model assigns $r_\theta(a_1) = 3$ and $r_\theta(a_2) = 1$, then the inferred probability that a_1 is preferred over a_2 is $\Pr(a_1 \succ a_2) = \frac{3}{3+1} = 0.75$.

Nevertheless, recent work has raised concerns about limitations of BT-style reward models. One issue is over-optimization and poor generalization when RMs are used for RLHF policy training; models may exploit artifacts rather than truthfully reflect human judgments [8, ?]. Another line of work argues that BT’s pairwise comparison assumption may be unnecessarily restrictive, especially when human preferences exhibit richness such as intransitivity or context dependence [3].

2.2 Alternative Reward Modeling techniques

Recent work has explored alternatives to embedding-based BT reward models, motivated by the goals of improving robustness, generalization, and handling richer preference phenomena.

Some work proposes preference models based on energy and contrastive measures; such models replace the logistic assumptions with an energy-based framework, trained via contrastive losses. This formulation allows for a unique maximum-likelihood solution, which mitigates issues of non-identifiability and yields more stable reward estimates [10]. Further, other work explores multi-objective or hybrid architectures, reflecting the insight that human judgments often depend on multiple criteria: helpfulness, coherence, safety, style, etc. Rather than collapsing all dimensions into one single scalar, these models learn separate sub-objectives and offer a structured, composite score, which potentially capture trade-offs that a scalar measure cannot [4, 28]. Finally, other works seek to address these challenges with emphasis on regularization, constraint enforcement, uncertainty estimations, or latent-structure modeling [14, 16, 24, 29, 20].

Beyond this, another growing field in this area removes the explicit modeling of rewards. With direct preference optimization (DPO), the algorithm directly fine-tunes the language model on raw human reference data. By training on annotated (prompt, chosen response, rejected response) data using a classification loss, DPO simplifies the alignment pipeline and avoids potential instabilities of RL [25].

In all, these alternative modeling strategies mark a shift away from the classic BT, monolithic, scalar reward models. Our proposed work takes inspiration from this landscape: by combining embedding representations, latent-factor structure, and strategic matrix imputation, we aim to contribute a new approach that improves generalization, robustness, and fidelity to human preferences.

2.3 Matrix factorization and Recommender Systems

Outside of LLM alignment, the community has developed many methods of modeling user-item interactions based on latent-factor models and matrix factorization (MF), especially in the context of recommender systems [1, 12, 18, 17]. The work is able to impute and approximate information in large, sparse user-item interaction matrices (e.g. ratings, clicks, implicit feedback) using low-dimensional latent user and item factor matrices. This is thus able to capture shared latent preferences and enabling generalization to unseen user-item pairs.

Continuing with these methods, recent work also have shown that we can combine item side information (such as embeddings of item content) with latent-factor models in recom-

mendation tasks. For example, methods that integrate word-based or content-based embeddings with MF can alleviate cold-start problems and improve recommendation accuracy [19]. Neural matrix-factorization variants also combine embedding-based features with latent interaction modeling [13].

3 Proposed Work

We propose to implement and evaluate a set of reward-modeling strategies, each combining human feedback, embeddings, and matrix / latent-factor methods.

3.1 Datasets

We make use of the following publicly available human-preference / alignment datasets:

- **Anthropic_HH_Golden** — a preference-label dataset extending the helpful/harmless (HH) alignment data [4].
- **HelpSteer2** — a dataset of human judgments on helpfulness/coherence for model outputs, designed explicitly for reward-model training [27].
- **Summarize_from_Feedback** — the feedback data released by the “Learning to Summarize from Human Feedback” effort: contains comparisons between candidate summaries for the same prompt [26].
- **Collective-Alignment-1** — a dataset of human preference comparisons under a collective-alignment initiative [22].
- **Prism-Alignment** — another open preference / alignment dataset for LLM outputs [15].

Notably, these datasets have various structures that we process in order to ensure they fit the same format. For Anthropic_HH_Golden and HelpSteer2, we treat the data as coming from an aggregate “population user” (i.e. no per-annotator / user-specific modeling), because persistent user (annotator) identifiers are unavailable. For Collective-Alignment-1 and PRISM, which have many-to-one or one-to-many preference formats (e.g. when one “chosen” response is compared against multiple “rejected” candidates), we convert them into pairwise comparisons. For example, when a prompt P has a chosen response R_{chosen} and multiple rejected responses $R_1, R_2, \dots$, we generate comparison pairs $(P, R_{\text{chosen}}, R_1), (P, R_{\text{chosen}}, R_2), \dots$ for inclusion in our (prompt, response_1, response_2, label) data set.

Notably, we also note that it could be interesting to process the data in a way that is domain explicit; for each prompt topic (e.g. "safety"), we can view that as coming from one "user" to see if the model better generalizes to capture these differences in content.

3.2 Evaluation Metrics

To evaluate reward-models, we will primarily use pairwise ranking accuracy, the current standard method of evaluating reward models in RLHF. This will be computed as the fraction of held-out comparisons where the model correctly predicts which response was preferred by humans. This will be computed over various parameters (including different train sizes, embedders, and other factors) over a series of 5 runs. This is to measure some measure of significance in our reward model accuracy scores.

Further, to evaluate generalization, especially the ability of a reward model to assess novel prompts or responses, we will examine performance on out-of-distribution (OOD) data splits. These OOD tests could involve prompts from a different domain, responses with different style/length, or responses from completely different datasets. The goal is to assess whether a model trained on a limited subset of feedback still produces reasonable scores when faced with new, varied inputs.

Finally, we will conduct qualitative error analysis and robustness diagnostics. This includes manually inspecting examples where models disagree with human preference, especially "edge cases" such as very long or rare responses, adversarial / low-quality responses, or subtle style/content differences. We will also evaluate the reward model further downstream in the RLHF pipeline, and track policy drift (via tracking distributional shift of generated outputs and divergence metrics between the policy and reference distribution).

3.3 Proposed Reward Models

We define our proposed reward models as follows (note the baseline we compare to is the Bradley-Terry model as the standard definition in the literature). Note that all require a training sample defined as comparisons (u, P, r_1, r_2, y) and an embedder that can construct these responses from language to vectors.

For the first three algorithms, we construct an imputed matrix whose rows correspond to latent preference entities (referred to as "users") and whose columns correspond to (prompt, response) pairs. As stated in 3.1, when explicit user identifiers are unavailable, we can both

consider treating the results coming from one user, or even creating latent entities that are instantiated by prompt domains.

Next, in order to create a score for a (P, R) pair, we can then compute the cosine similarity between the imputed matrix from the algorithm, then use it to impute a new row which we can average across all users.

Algorithm 1 Simple ME (Unary) Model

1. Construct a matrix S with user rows and (P, R) cols

Initialize dictionaries: $\text{wins}[u, P, r] = 0$, $\text{games}[u, P, r] = 0$

for all comparisons (u, P, r_w, r_l, y) in training data **do**

$\text{wins}[u, P, r_w] += y$; $\text{games}[u, P, r_w] += 1$

$\text{wins}[u, P, r_l] += (1 - y)$; $\text{games}[u, P, r_l] += 1$

end for

for all (u, P, r) with $\text{games}[u, P, r] > 0$ **do**

Embed (P, r) to get vector e

$s(u, e) \leftarrow \frac{\text{wins}[u, P, r]}{\text{games}[u, P, r]}$

end for

Apply SVD++ with item embeddings to fully impute S

return S

Algorithm 2 ME Pairwise-to-Response Model

1. Build a matrix with the data given

Build user-pairwise matrix over triplets (P, r_1, r_2)

Impute missing pairwise preferences via SVD++ to obtain completed $\widehat{M}$

2. Construct a matrix S as user-(prompt, response) as follows

for all triplets (P, r_1, r_2) **do**

for all users u **do**

$p \leftarrow \widehat{M}[u, (P, r_1, r_2)]$

$\triangleright$ imputed $P(r_1 \succ r_2)$

if $S[u, P, r_1]$ and $S[u, P, r_2]$ are both unset **then**

 Set $S[u, P, r_1] \leftarrow p$, $S[u, P, r_2] \leftarrow 1 - p$

else if $S[u, P, r_1]$ unset, $S[u, P, r_2]$ set (and vice versa) **then**

$S[u, P, r_1] \leftarrow S[u, P, r_2] \cdot \frac{p}{1 - p}$

else

 Let current ratio $R_{\text{rel}} \leftarrow \frac{S[u, P, r_1]}{S[u, P, r_2]}$

 Let new ratio $R_{\text{new}} \leftarrow \frac{p}{1 - p}$

if R_{rel} and R_{new} disagree in direction **then**

 Reset all $S[u, P, \cdot]$ for this prompt P to be uniform over its responses

else

 Update S : $\max(R_{\text{rel}}, R_{\text{new}}) = \frac{S[u, P, r_1]}{S[u, P, r_2]}$ with $S[u, P, r_1] + S[u, P, r_2]$ fixed

end if

end if

end for

end for

Normalize $\{S[u, P, r]\}_r$ so that $\sum_r S[u, P, r] = 1$

return S

Algorithm 3 ME Score-Diff MF Model

1. Train a pairwise MF model

Create a user-item MF model that predicts a score $s(u, P, r)$.

Embed (P, r) and use as features to guide the MF factors.

Train using logistic loss on the score difference:

$$s(u, P, r_1) - s(u, P, r_2) \approx \log \frac{y}{1 - y}.$$

2. Produce a complete score matrix

After training, compute the dense matrix:

$$S[u, (P, r)] = s(u, P, r)$$

for all users and items.

return S

Algorithm 4 ME Diff Pairwise-Baseline Model

1. Construct difference vectors**for all** comparisons (u, P, r_1, r_2, y) **do** Embed (P, r_1) and (P, r_2) to get e_1 and e_2 . If $y \geq 0.5$, define difference $d = e_1 - e_2$; else $d = e_2 - e_1$. Store (u, d) as a training example meaning “good > bad”.**end for****2. Fit a global difference model**

Learn a linear scorer:

$$\text{score}(u, d) = w^\top d + b_u$$

3. Build a replay bufferCollect a set of B baseline responses drawn from the training data.

Embed each baseline and store

4. Scoring a new (P,r)Embed (P, r) to get e^* .For each baseline b_j , compute a difference vector:

$$d_j = e^* - b_j.$$

For each user u , predict probabilities $\Pr_u(e^* \succ b_j)$.

Average across baselines and users:

$$R(P, r) = \frac{1}{|U|B} \sum_{u,j} \Pr_u(e^* \succ b_j).$$

return $R(P, r)$

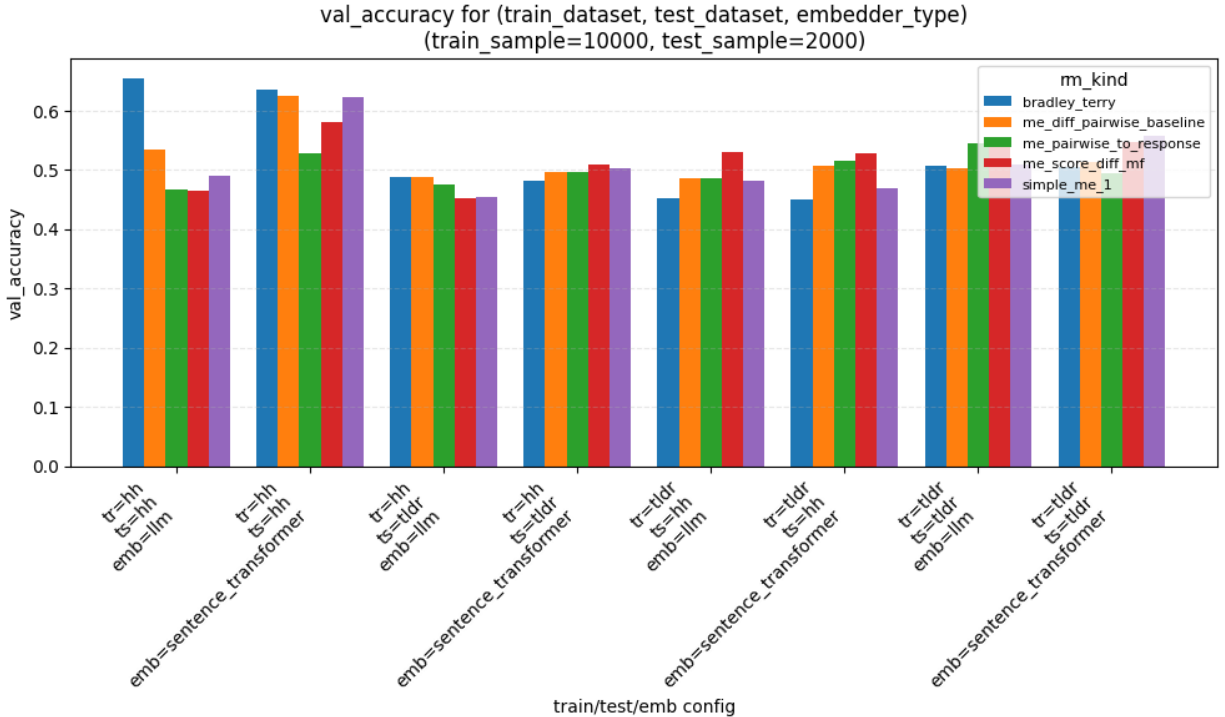


Figure 1: Reward models evaluated across different train and test sets, as well as embedders.

With these algorithms, we have some preliminary results as seen in Figure 1. We can see varying levels of performance in our new reward models, indicating there is some promise, but still exploration to be done.

3.3.1 Other Potential Avenues of Exploration

Beyond the four reward models defined above, we can further explore other directions within this space.

First, a major point of interest is the quality and structure of the underlying embeddings. One avenue is a systematic study of different sentence embedders (e.g., multiple sizes and architectures of recent sentence-transformer models) and how their geometry behaves under this matrix estimation and cosine-similarity-based scoring. This could include preprocessing of these embeddings to help us discover or engineer a low-rank or sparse + low-rank formulation, as well as generalizing how different sizes of these sentence embeddings affect performance. Some ideas for preprocessing are clipping, calibration, or other techniques to allow these embeddings to be more generalizable. Further, we could potentially investigate if jointly fine-tuning the embedder with the reward model can enforce more structured pref-

erence matrices.

Second, it would be interesting to go beyond purely outcome-based pairwise labels, and incorporate richer supervision signals. For example, chain-of-thought (CoT) work, process-based feedback, or GRPO-style preference data could be treated as additional views of the same underlying reward and analyzed through the same matrix estimation lens. Further, we can explore credit assignment in the reward function; we know that the prompts are treated token by token, yet the reward is assigned to the entire prompt without regards to which was the most beneficial.

Third, there is an opportunity to further study the integration of data between users and domains. We can explore MOE architectures in this space, essentially aiming to capture this idea of clusters that may exist in the data. Finally, we could compare training a single unified reward model over various datasets versus training on domain-specific ones with this matrix estimation framework.

3.4 Timeline

Month	Planned Milestones
Dec	Process all datasets and confirm baseline BT performance. Conduct NeurIPS + CDC literature review. Implement additional variants (reward clipping, sparse + low-rank ME, alt similarity metrics beyond cosine similarity). Run experiments across new RMs (vary train size, datasets, embedders). Review sentence-transformer architectures and embedding literature.
Jan	Test additional embedders Add preprocessing/calibration to embedders induce or reveal latent structure. Continue modifying reward models as ideated. [ICML abstract deadline (Jan 23)].
Feb	Implement and test MOE-style RM models, integrated with matrix structure Explore multi-dataset training for generalizability.
Mar	Move beyond scalar reward functions: read on CoT, process supervision, GRPO/ORPO, PBRL, credit assignment, etc. Implement 1–2 reasoning-focused RM baselines for comparison.
Apr	Extend reasoning-aware reward modeling using matrix estimation ideas. Explore: “How would reasoning appear as matrix structure?”; analyze current approaches. Conduct ICLR literature review.
May	Explore interpretability; why do the models behave as they do? [NeurIPS abstract deadline (mid May)]. [EMNLP abstract deadline (mid May)].
Jun	Study integration of LLMs and the RLHF pipeline (ie. DPO). Evaluate whether ME-style RMs can complement or replace DPO-style pipelines.
Jul	Continue exploration of DPO and non-RL alignment methods Explore theoretical considerations: limits of scalar rewards, structure in preference data, scalability of reward models. Conduct ICML + ACL literature review. [AAAI abstract deadline (late July)].
Aug	Explore alternative RLHF training paradigms (REINFORCE variants, hyperspecific RL algorithms for LLMs). Explore of diffusion-based LLMs (and potentially how would a reward model fit there in RLHF?).
Sep–Oct	Wrap up any ideas that came up on the prior topics, exploratory work! [ICLR abstract deadline (mid Sep)].
Nov–Dec	Wrap up any remaining ideas, consolidate results, and write thesis.

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